Divya C Nambiar

Senior Cloud AI/ML Architect — Hewlett Packard Enterprise (HPE)

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Professional Profile

Senior Cloud AI/ML Architect with a proven record of designing and scaling production LLM/agentic systems, predictive analytics pipelines, and time-series foundation-model forecasting. Passionate about applying deep technical expertise and inventive problem-solving to drive innovation on cutting-edge, research-driven products.

Professional Summary

- 16+ years in software development
- 8+ years in Data Science / ML
- Python, FastAPI, PySpark, SQL, MLOps
- LLMs, RAG, and classical ML algorithms
- Multi-LLM orchestration, routing & benchmarking
- Time-series foundation-model forecasting at scale
- Predictive analytics for hardware reliability
- 9 patents/IP filings and 2 trede secrets across LLM, forecasting, Log Analytics& federated learning

Patents & Innovation

Inventor and technical lead on 9 patents/2 trade secret filings; each drove a shipped product feature.

- **Auto-Tuned Swarm Learning Models** — *Granted, US Patent*.
- **Auto-Merging Decentralized-Training Metadata in Machine Learning Tracking Pipeline** — *Published application (pending)*.
- **Policy-Driven Selective Metadata Synchronization in CMF Meta-Hub Using Meta-Card Attributes** — selective, attribute-based metadata sync to cut redundant transfer across hubs. *Filed.*
- **Adaptive Context Optimization for Long-Horizon Forecasting in Large-Scale Time Series Foundation Models** — primary author; anchoring-based method improving accuracy of large-scale time-series foundation models over long forecast horizons. *Filed; submitted to ACM CODS 2026.*
- **CompactRAG: Criticality-Aware Tiered Compression of Vector Embeddings for Retrieval-Augmented Generation Systems** — shrinks vector-store footprint while preserving retrieval quality. *Filed.*
- **End-to-End Feasibility Certification Pipeline for Constraint-Governed Model Serving** — pipeline certifying models meet serving constraints before deployment. *Filed.*
- **AI-Driven Hardware Failure Prediction** — predicts CPU and DIMM failures ahead of occurrence from consolidated telemetry data. *Filed; formal application in progress.*
- **Log-ifying Timeseries Data for Unified Observability and LLMbased Root Cause Analysis** — *HPE Trade Secret*
- **Cross-dataset meta-learning for metric forecasting with limited historical data** — *HPE Trade Secret*

Work Experience

Hewlett Packard Enterprise (HPE) | Senior Technical Specialist / Cloud AI-ML Architect — *Dec 2015 – Present*
Hewlett Packard Software (Now Open Text) | Senior Software Developer — *Sep 2011 – Dec 2015*
Tecknodreams Software Consulting Pvt Ltd | Senior Software Developer — *Dec 2010 – Sep 2011*

Key Projects

Enterprise AI Co-Pilot for Observability — HPE Compute Ops Management (COM) *Ongoing*

- Architecting and scaling an LLM-powered co-pilot integrated into the observability platform, from initial rollout through production growth.
- Designed a multi-LLM routing and benchmarking framework that automatically selects the best-fit model per use case on accuracy, latency, and cost — filed as HPE IP.
- *Leadership:* Leading a new investigation into training a custom LLM on operational log data for automated root-cause analysis.

AI-Driven Hardware Failure Prediction — HPE Compute Ops Management (COM) *Recent*

- Led the end-to-end POC and implementation, shifting server-health monitoring from reactive to proactive failure-risk insights.
- Consolidated fragmented telemetry and inventory data from multiple business units into a unified training dataset.
- Built GPU-accelerated training and inference pipelines for CPU and DIMM failure prediction.
- *Achievement:* Filed as HPE IP; accepted for HPE TechCon 2027.

Sustainability Metrics Forecasting with Foundation Models — HPE Compute Ops Management (COM) *Recent*

- Led the end-to-end production implementation, migrating from independently trained, per-server Prophet models to Google TimesFM for scalable group-level power-utilization forecasting.
- Led the project; primary author on two inventions solving open forecasting problems: low-data forecasting and long-horizon forecasting via anchoring.
- *Achievement:* Long-horizon anchoring method filed as HPE IP and submitted to ACM CODS 2026.

HPE Green Lake Log Analytics *Apr 2024 – Present*

- Built FastAPI services to ingest CSV and Kafka input streams.
- Applied LangChain + LLMs to cluster and analyze logs for root-cause insights.
- Designed UI templates integrated with the FastAPI backend.
- *Leadership:* UI design & POC; secondary author of the business proposal (HPE TechCon).
- *Achievement:* One of the best business ideas — HPE TechCon 2023.

HPE Swarm Learning *Mar 2022 – Feb 2024*

- Designed and implemented federated MLOps for Swarm Learning.
- Built custom models with Keras and PyTorch for decentralized training.
- Architected CI/CD and led development across multiple personas.
- *Leadership:* Led CI/CD, UI features, and Swarm–Determined AI integration.
- *Achievement:* Multiple team recognitions for the Swarm Learning GA release.

HPE OneView *Jan 2016 – Feb 2022*

- Created an NLP-based duplicate-ticket finder.
- Troubleshot customer cases and defects; authored complex PostgreSQL queries.
- *Leadership:* Networking Adapter lead; WoW Branch Master member.
- *Achievement:* "Obsess over the customer" & "Push for better" recognitions; multiple Branch Master awards.

Early Career — HPE Software & Tecknodreams *2010 – 2015*

- Virtualization Performance Viewer: content packs and DB stored procedures to classify under/over-sized VMs on column-store databases (HP Vertica, Sybase).
- Restructured Accelerated Power Development: Java web services with stored procedures, functions, and triggers.

Education

BITS Pilani — PG in AI/ML (CGPA 9.4, Distinction) *2018*
CUSAT, Kerala — MCA (80%, Distinction) *2008*
Kannur University, Kerala — BSc Computer Science, Maths and Statistics (90%, Distinction, 5th Rank) *2005*

Certifications

- Databricks (Coursera)
- Spark, Hadoop & Snowflake for Data Engineering (Coursera)
- Docker (HPE)
- MEAN Stack (HPE)
- SAFe Agile / Agile (HPE)
- Advanced Python (HPE)

Leadership

- Enabled team members to contribute to feature-based IP submissions and gain recognition for their work.
- Led the COM SMS-level hackathon, earning recognition for cross-team collaboration.
- Mentored university-relations participants and junior engineers pursuing M.Tech degrees.
- Recognized by stakeholders for technical input that supported informed decision-making.

Recognition & Awards

- Selected presenter, **HPE TechCon 2026** (poster); work also earned an **Honorable Mention** at HPE Regional TechCon 2026.
- Certified **HPE TechCon / IP Reviewer**.
- Multi-time **Star of the Month** awardee (2025–2026) for leadership on hardware failure prediction and foundation-model forecasting delivery.